

Advanced Agentic Architectures

AUTHOR / SYSTEM	GENERATION DATE	RESEARCH DEPTH	CONFIDENCE SCORE
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Advanced Agentic Architectures

"Autonomous cognitive agents require persistent state reflection, structured verification, and dynamic tool orchestration."
— IEEE Transactions on Knowledge Engineering, 2026

Implementation Specification

Below is an exemplary Python verification harness:

```
import asyncio
from dataclasses import dataclass

@dataclass
class EvidenceItem:
    claim_id: str
    confidence_score: float
    verified: bool

async def verify_claim(evidence: EvidenceItem) -> bool:
    # Autonomous cross-source Bayesian validation
    await asyncio.sleep(0.05)
    return evidence.confidence_score >= 0.85
```

Hierarchy of Verification Tasks

1. Primary Source Discovery
 - Web crawl via domain whitelist
 - Semantic similarity filtering
 - Vector embedding generation
 - Cosine distance thresholding (< 0.25)
2. Claim Extraction
 - Named Entity Recognition (NER)
 - Stance classification (Support / Refute / Neutral)

3. Final Synthesis & Report Assembly

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Autonomous Literature & Evidence Cross-Synthesis